# Customer Churn Prediction and Agentic Retention Strategy

Project 5
B.Tech CSE (AI)

**Abstract**

This project implements a complete customer churn analytics system in two milestones. Milestone 1 delivers a machine learning pipeline for telecom churn prediction using classical supervised learning, preprocessing, model evaluation, visualization, and an interactive Gradio interface. Milestone 2 extends the same system into an agentic retention strategist built with LangGraph, local retrieval using FAISS, and Gemini-based structured generation. The final system predicts churn risk, explains the major churn drivers, retrieves grounded retention playbooks, and generates actionable intervention plans with follow-up question answering.

## 1 Problem Statement

Customer churn causes direct revenue loss and increases acquisition cost for telecom businesses. The objective of this project is to:

- predict whether a customer is likely to churn,
- identify the main business drivers of that risk,
- present both single-customer and batch predictions through a hosted application,
- extend the prediction system into an AI assistant that proposes retention actions.

## 2 Dataset Description

The project uses the Telco Customer Churn dataset containing 7,043 customer records. Each row represents one customer and includes demographic, service, and billing information.

Important numeric features:

- tenure
- MonthlyCharges
- TotalCharges

Important categorical features:

- gender
- Partner
- Dependents
- InternetService
- Contract
- PaymentMethod
- TechSupport
- OnlineSecurity

The target variable `Churn` is encoded as:

- 1 for churn
- 0 for retained

# 3 Milestone 1: ML-Based Churn Prediction

## 3.1 Data Cleaning and Preprocessing

The preprocessing pipeline is implemented using `ColumnTransformer`.

Cleaning steps:

- convert `TotalCharges` to numeric,
- impute missing numeric values,
- remove the customer identifier,
- encode the churn label,
- standardize numeric features,
- one-hot encode categorical features.

## 3.2 Exploratory Data Analysis

The EDA module performs:

- missing-value checks,
- churn distribution analysis,
- skewness checks,
- outlier detection,
- multicollinearity analysis using VIF,
- correlation heatmap generation.

## 3.3 Model Development

Two production models are used in the hosted system:

- Logistic Regression
- Decision Tree

The Logistic Regression pipeline includes polynomial interaction features, Lasso-based feature selection, and SMOTE balancing. The Decision Tree pipeline uses preprocessing, SMOTE, and tuned tree hyperparameters.

## 3.4 Evaluation Metrics

The models are evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Table 1: Final Test Set Performance

| Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|
| Logistic Regression | 79.2% | 59.4% | 68.5% | 63.6% |
| Decision Tree | 74.7% | 51.6% | 76.4% | 61.6% |

The Logistic Regression model is used as the default scoring model for the agentic workflow because it offers strong overall performance and clearer driver explanations. The Decision Tree remains available for comparison inside the dashboard.

# 4 Milestone 1 Interface

The hosted Gradio application provides:

- model performance dashboard,
- EDA visualizations,
- single-customer prediction form,
- CSV batch scoring with downloadable results,
- top churn-driver summaries.

# 5 Milestone 2: Agentic Retention Strategist

## 5.1 System Overview

Milestone 2 extends the prediction system into a stateful agent that turns churn risk into retention recommendations. The workflow is implemented with LangGraph and uses a local knowledge base plus retrieval.

## 5.2 Agent Workflow

The LangGraph flow contains the following stages:

1. prepare customer input,
2. score churn risk using the shared inference layer,
3. summarize churn drivers and business context,
4. retrieve relevant retention strategies from FAISS,
5. generate a structured retention report using Gemini,
6. answer follow-up questions using the same case state.

## 5.3 Explicit State

The graph stores:

- customer payload,
- chosen model,
- prediction result,
- risk summary,

- retrieval query,
- retrieved evidence,
- structured retention report,
- follow-up conversation history.

## 5.4 Retrieval-Augmented Generation

The knowledge base contains telecom-specific playbooks for:

- pricing retention,
- contract upgrades,
- support recovery,
- payment friction,
- digital adoption,
- high-value customer handling.

Documents are embedded using `gemini-embedding-001` and indexed locally with FAISS. This keeps the system compatible with free-tier hosting constraints.

## 5.5 Structured Output

The retention strategist returns a report with:

- business context,
- risk summary,
- key drivers,
- retrieved evidence,
- recommended actions,
- priority order,
- next-touch plan,
- confidence notes.

# 6 Architecture Summary

1. User submits a customer profile or CSV file.
2. Shared inference loads the trained ML artifacts and computes churn probability.
3. The app extracts major churn drivers for explainability.
4. The LangGraph workflow formulates a retrieval query.
5. FAISS returns relevant retention playbooks.
6. Gemini generates a structured retention plan grounded in the retrieved evidence.
7. The user can ask follow-up questions inside the same case session.

# 7 Deployment

The final system is deployed as a hosted Gradio application on Hugging Face Spaces. The deployment uses free-tier compatible components:

- Gradio for the interface,
- scikit-learn for ML inference,
- LangGraph for agent orchestration,
- FAISS for local retrieval,
- Gemini API for structured report generation.

The app reads `GEMINI_API_KEY` from Space secrets. When model artifacts are absent in a deployment branch, the application can bootstrap training automatically before serving predictions.

# 8 Visual Results

## 8.1 ROC Curve

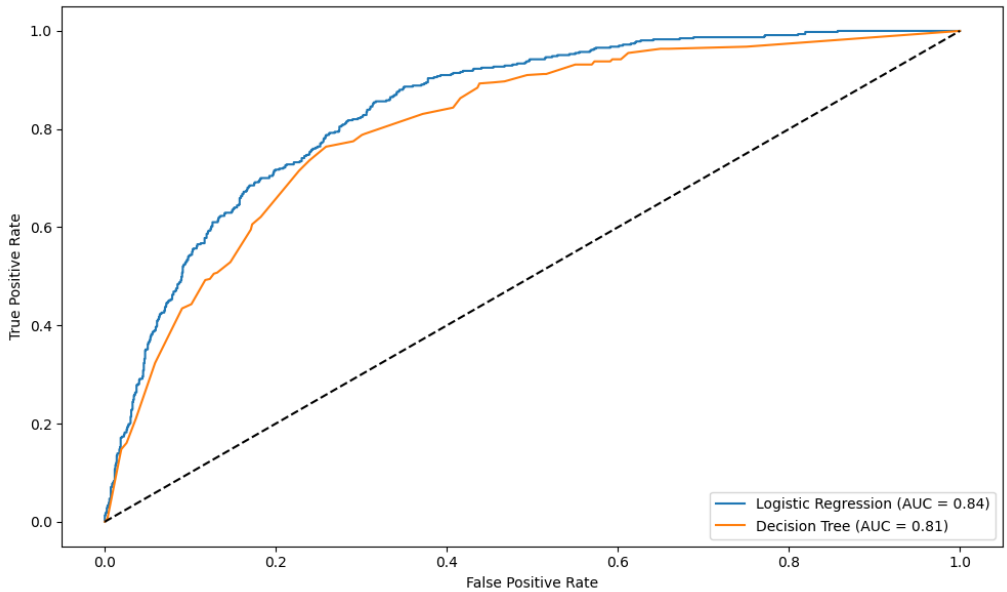


Figure 1: ROC curve comparison for the deployed models

## 8.2 Precision-Recall Curve

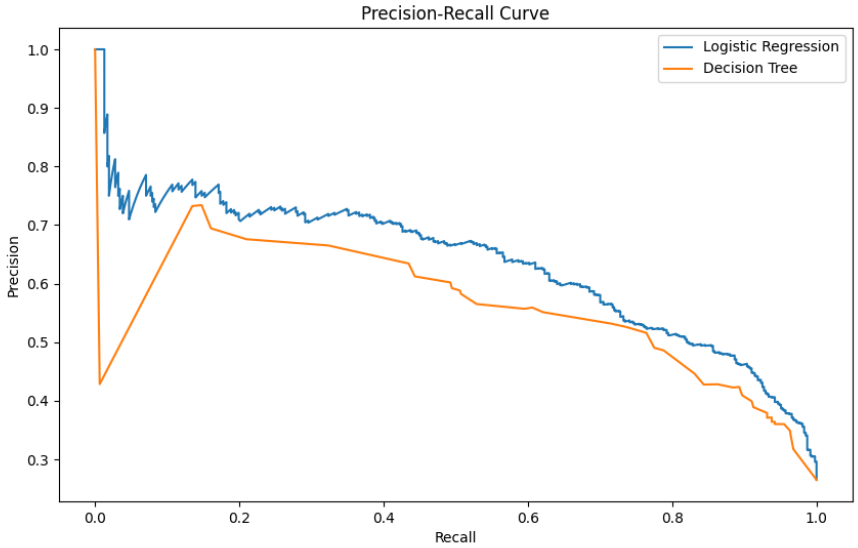


Figure 2: Precision-recall curve comparison

## 8.3 Logistic Regression Confusion Matrix

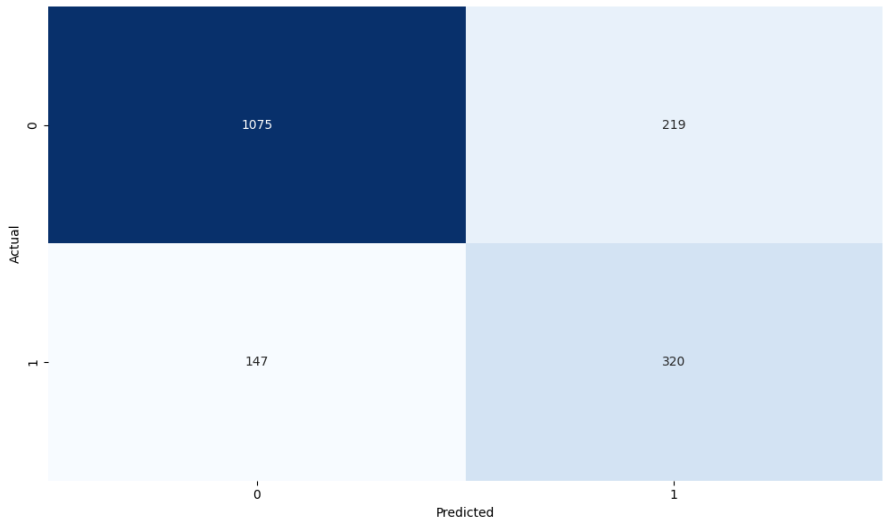


Figure 3: Logistic Regression confusion matrix

## 8.4 Decision Tree Confusion Matrix

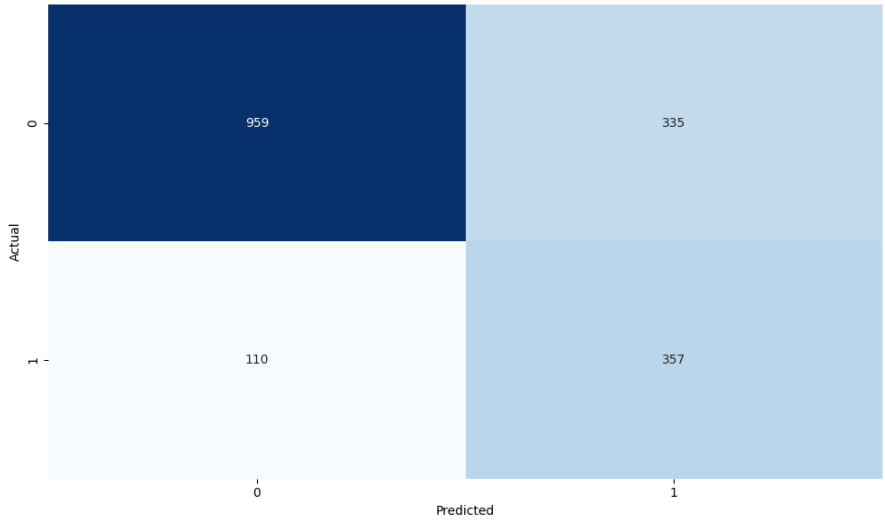


Figure 4: Decision Tree confusion matrix

## 8.5 Logistic Regression Driver Coefficients

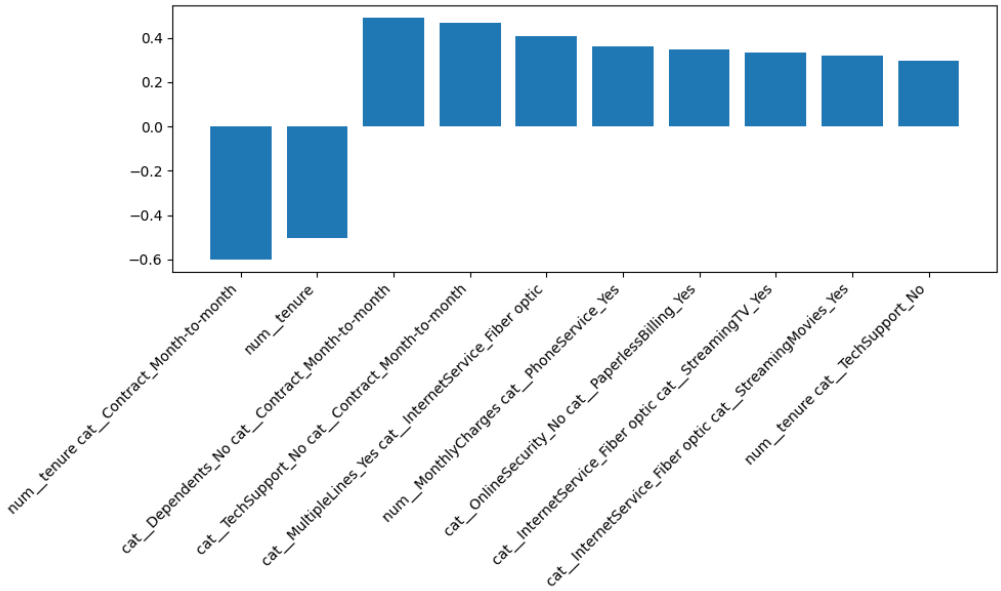


Figure 5: Top Logistic Regression coefficients

## 8.6 Decision Tree Feature Importance

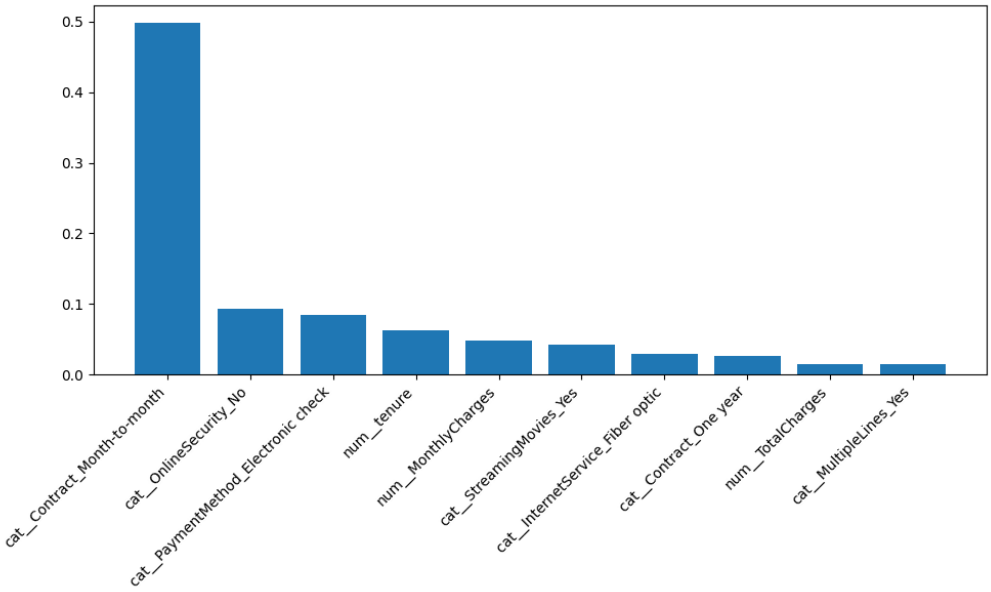


Figure 6: Decision Tree feature importance

## 8.7 EDA: Tenure Distribution

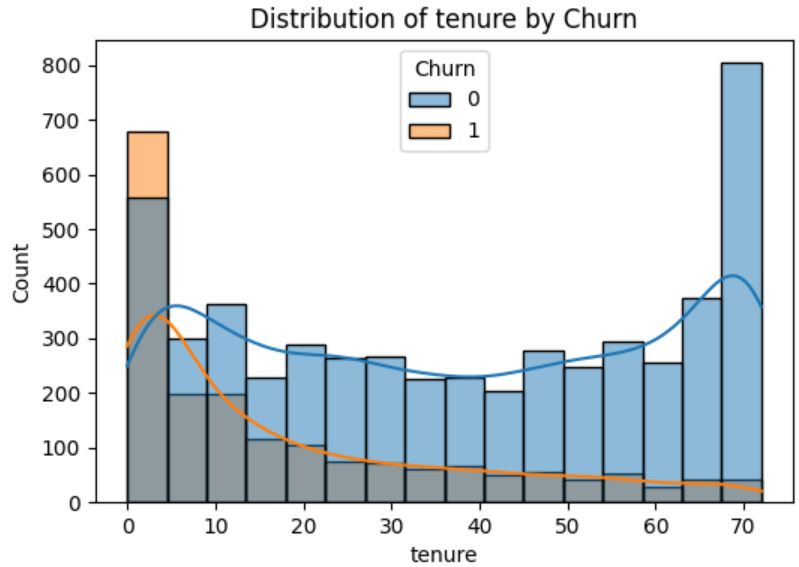


Figure 7: Tenure distribution by churn class

## 8.8 EDA: Monthly Charges Distribution

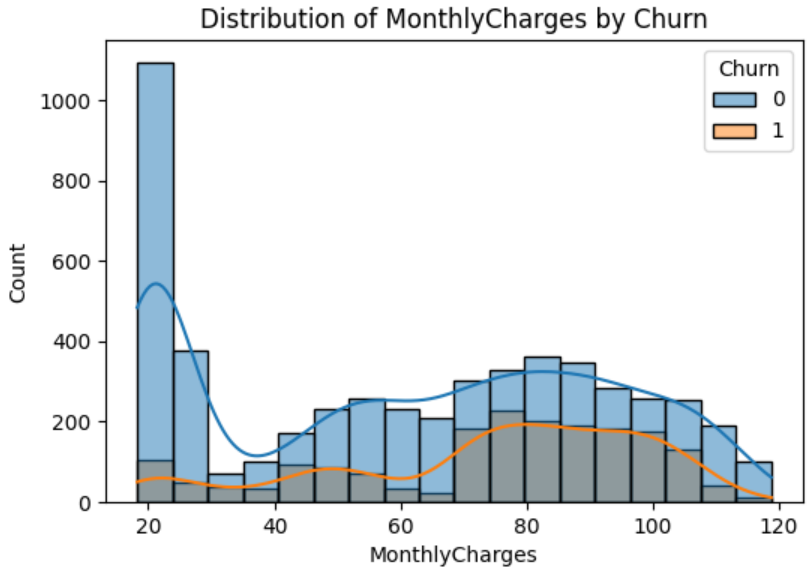


Figure 8: Monthly charges distribution by churn class

## 8.9 EDA: Total Charges Distribution

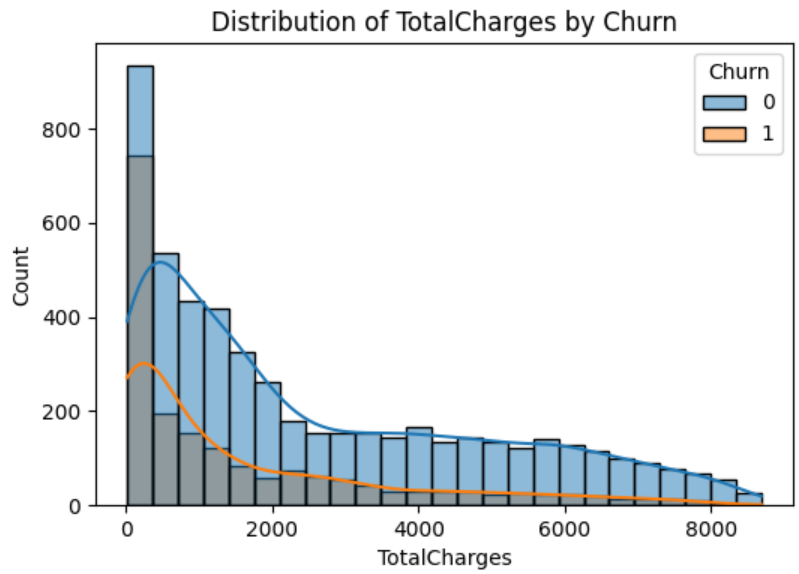


Figure 9: Total charges distribution by churn class

## 8.10 EDA: Correlation Matrix

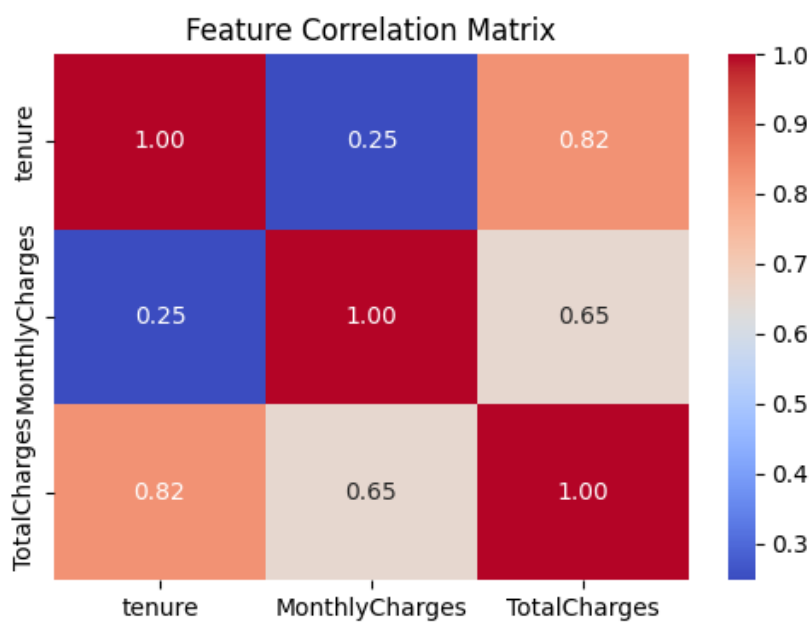


Figure 10: Correlation matrix for numeric features

# 9 Conclusion

This project demonstrates a full progression from predictive analytics to intelligent intervention. Milestone 1 provides a robust churn prediction system with explainable outputs and hosted usability. Milestone 2 adds an agentic retention layer that reasons over churn risk, retrieves grounded best practices, and produces structured actions for business teams. The resulting system satisfies both the predictive ML requirements and the agentic AI retention strategy requirements of the project.

# 10 References


- Scikit-Learn Documentation
- Imbalanced-Learn Documentation
- LangGraph Documentation
- FAISS Documentation
- Google Gemini API Documentation